CLASS X : CHAPTER - 2
POLYNOMIALS

IMPORTANT FORMULAS & CONCEPTS

An algebraic expression of the form $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots\ldots\ldots\ldots a_nx^n$, where $a \neq 0$, is called a polynomial in variable x of degree n.
Here, $a_0, a_1, a_2, a_3, \ldots\ldots\ldots, a_n$ are real numbers and each power of x is a non-negative integer.
e.g. $3x^2 - 5x + 2$ is a polynomial of degree 2.
$3\sqrt{x} + 2$ is not a polynomial.

- If $p(x)$ is a polynomial in x, the highest power of x in $p(x)$ is called **the degree of the polynomial** $p(x)$. For example, $4x + 2$ is a polynomial in the variable x of degree 1, $2y^2 - 3y + 4$ is a polynomial in the variable y of degree 2,

  - A polynomial of degree 0 is called a constant polynomial.
  - A polynomial $p(x) = ax + b$ of degree 1 is called a linear polynomial.
  - A polynomial $p(x) = ax^2 + bx + c$ of degree 2 is called a quadratic polynomial.
  - A polynomial $p(x) = ax^3 + bx^2 + cx + d$ of degree 3 is called a cubic polynomial.
  - A polynomial $p(x) = ax^4 + bx^3 + cx^2 + dx + e$ of degree 4 is called a bi-quadratic polynomial.

VALUE OF A POLYNOMIAL AT A GIVEN POINT x = k
If $p(x)$ is a polynomial in x, and if k is any real number, then the value obtained by replacing x by k in $p(x)$, is called **the value of $p(x)$ at $x = k$**, and is denoted by $p(k)$.

ZERO OF A POLYNOMIAL
A real number k is said to be a **zero of a polynomial $p(x)$**, if $p(k) = 0$.

- Geometrically, the zeroes of a polynomial $p(x)$ are precisely the x-coordinates of the points, where the graph of $y = p(x)$ intersects the x -axis.
- A quadratic polynomial can have at most 2 zeroes and a cubic polynomial can have at most 3 zeroes.
- In general, a polynomial of degree 'n' has at the most 'n' zeroes.

RELATIONSHIP BETWEEN ZEROES & COEFFICIENTS OF POLYNOMIALS

Type of Polynomial	General form	No. of zeroes	Relationship between zeroes and coefficients
Linear	$ax + b$, $a \neq 0$	1	$k = -\frac{b}{a}$, *i.e.* $k = -\frac{\text{Constant term}}{\text{Coefficient of x}}$
Quadratic	$ax^2 + bx + c$, $a \neq 0$	2	Sum of zeroes $(\alpha + \beta) = -\frac{\text{Coefficient of x}}{\text{Coefficient of x}^2} = -\frac{b}{a}$ Product of zeroes $(\alpha\beta) = \frac{\text{Constant term}}{\text{Coefficient of x}^2} = \frac{c}{a}$
Cubic	$ax^3 + bx^2 + cx + d$, $a \neq 0$	3	Sum of zeroes $(\alpha + \beta + \gamma) = -\frac{\text{Coefficient of x}^2}{\text{Coefficient of x}^3} = -\frac{b}{a}$ Product of sum of zeroes taken two at a time $(\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{\text{Coefficient of x}}{\text{Coefficient of x}^3} = \frac{c}{a}$ Product of zeroes $(\alpha\beta\gamma) = -\frac{\text{Constant term}}{\text{Coefficient of x}^3} = -\frac{d}{a}$

❖ A quadratic polynomial whose zeroes are α and β is given by $p(x) = x^2 - (\alpha + \beta)x + \alpha\beta$
i.e. x^2 – (Sum of zeroes)x + (Product of zeroes)

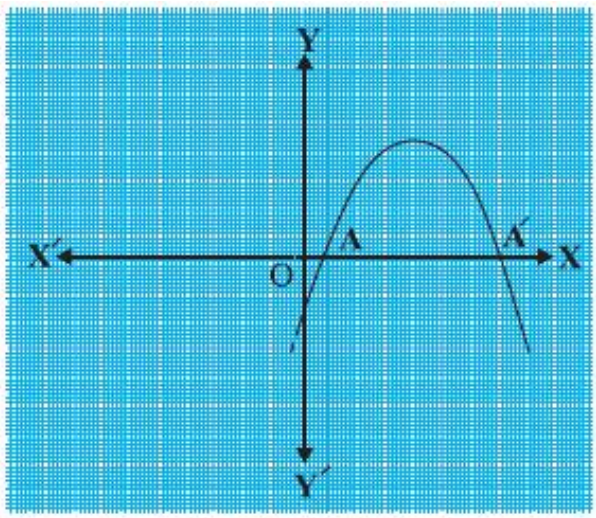
❖ A cubic polynomial whose zeroes are α, β *and* γ is given by
$p(x) = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma$

The zeroes of a quadratic polynomial $ax^2 + bx + c$, $a \neq 0$, are precisely the x-coordinates of the points where the parabola representing $y = ax^2 + bx + c$ intersects the x-axis.

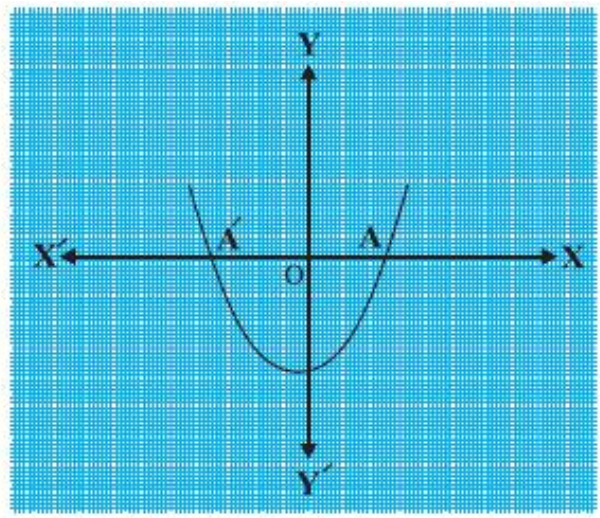
In fact, for any quadratic polynomial $ax^2 + bx + c$, $a \neq 0$, the graph of the corresponding equation $y = ax^2 + bx + c$ has one of the two shapes either open upwards like $\cup$ or open downwards like $\cap$ depending on whether $a > 0$ or $a < 0$. (These curves are called **parabolas**.)

The following three cases can be happen about the graph of quadratic polynomial $ax^2 + bx + c$:

Case (i) : Here, the graph cuts x-axis at two distinct points A and A'. The x-coordinates of A and A' are the **two zeroes** of the quadratic polynomial $ax^2 + bx + c$ in this case

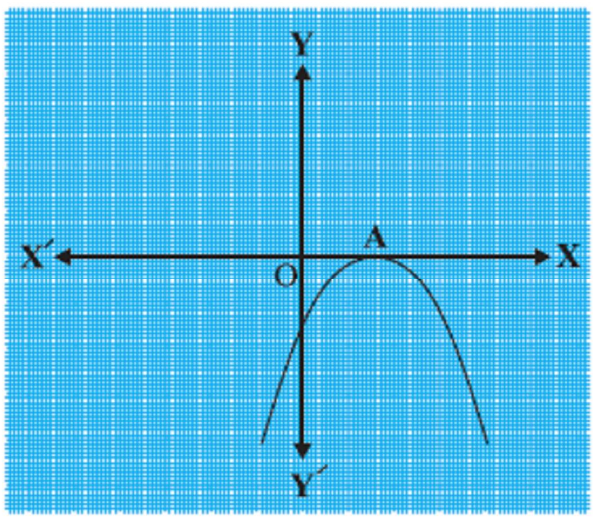


(i)
a > 0

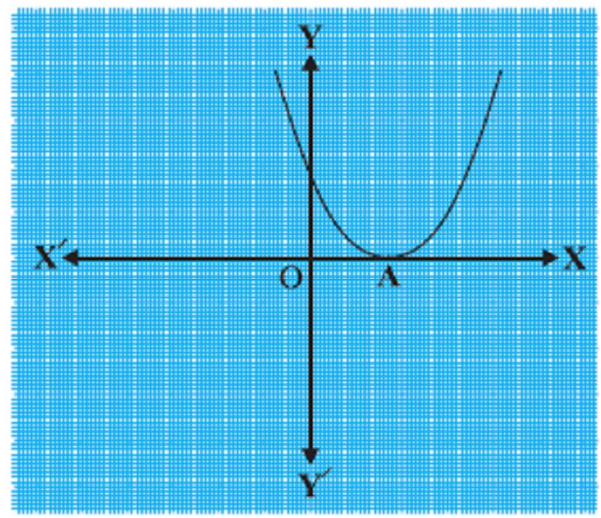


(ii)
a < 0

Case (ii) : Here, the graph cuts the x-axis at exactly one point, i.e., at two coincident points. So, the two points A and A′ of Case (i) coincide here to become one point A. The x-coordinate of A is the **only zero** for the quadratic polynomial $ax^2 + bx + c$ in this case.

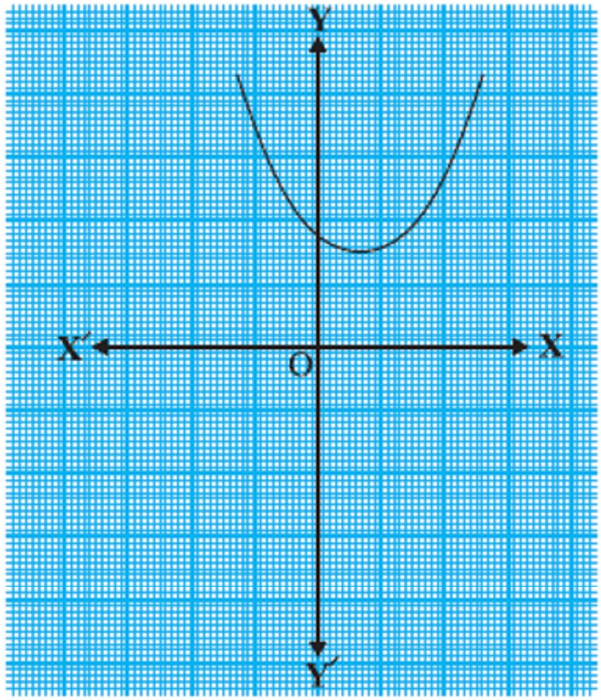


(i)
a > 0

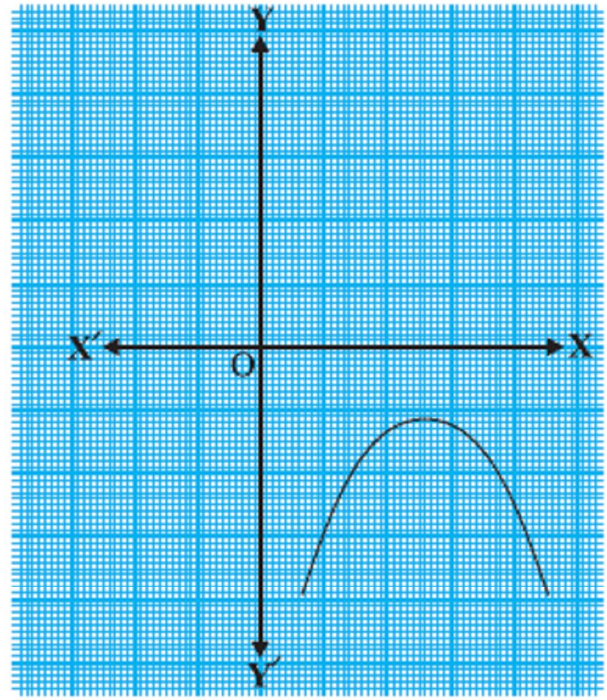


(ii)
a < 0

Case (iii) : Here, the graph is either completely above the *x*-axis or completely below the *x*-axis. So, it does not cut the *x*-axis at any point. So, the quadratic polynomial $ax^2 + bx + c$ has **no zero** in this case.



(i)
a > 0



(ii)
a < 0

DIVISION ALGORITHM FOR POLYNOMIALS

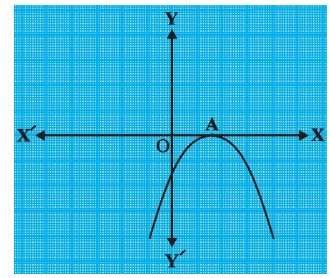
If $p(x)$ and $g(x)$ are any two polynomials with $g(x) \neq 0$, then we can find polynomials $q(x)$ and $r(x)$ such that $p(x) = g(x) \times q(x) + r(x)$,
where $r(x) = 0$ or degree of $r(x)$ < degree of $g(x)$.

- If r(x) = 0, then g(x) is a factor of p(x).
- Dividend = Divisor × Quotient + Remainder

MCQ WORKSHEET-I
CLASS X : CHAPTER - 2
POLYNOMIALS

1. The value of k for which (–4) is a zero of the polynomial $x^2 – x – (2k + 2)$ is
   (a) 3 (b) 9 (c) 6 (d) –1

2. If the zeroes of the quadratic polynomial ax2 + bx + c, $c \neq 0$ are equal, then
   (a) c and a have opposite sign (b) c and b have opposite sign
   (c) c and a have the same sign (d) c and b have the same sign



3. The number of zeroes of the polynomial from the graph is
   (a) 0 (b) 1 (c) 2 (d) 3

4. If one of the zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is
   (a) 10 (b) –10 (c) 5 (d) –5

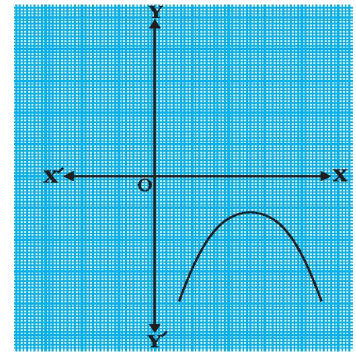
5. A quadratic polynomial whose zeroes are –3 and 4 is
   (a) $x^2 – x + 12$ (b) $x^2 + x + 12$ (c) $2x^2 + 2x – 24$. (d) none of the above.

6. The relationship between the zeroes and coefficients of the quadratic polynomial $ax^2 + bx + c$ is (a) $\alpha + \beta = \frac{c}{a}$ (b) $\alpha + \beta = \frac{-b}{a}$ (c) $\alpha + \beta = \frac{-c}{a}$ (d) $\alpha + \beta = \frac{b}{a}$

7. The zeroes of the polynomial $x^2 + 7x + 10$ are
   (a) 2 and 5 (b) –2 and 5 (c) –2 and –5 (d) 2 and –5

8. The relationship between the zeroes and coefficients of the quadratic polynomial $ax^2 + bx + c$ is (a) $\alpha.\beta = \frac{c}{a}$ (b) $\alpha.\beta = \frac{-b}{a}$ (c) $\alpha.\beta = \frac{-c}{a}$ (d) $\alpha.\beta = \frac{b}{a}$

9. The zeroes of the polynomial $x^2 – 3$ are
   (a) 2 and 5 (b) –2 and 5 (c) –2 and –5 (d) none of the above



10. The number of zeroes of the polynomial from the graph is
    (a) 0 (b) 1 (c) 2 (d) 3

11. A quadratic polynomial whose sum and product of zeroes are –3 and 2 is
    (a) $x^2 – 3x + 2$ (b) $x^2 + 3x + 2$ (c) $x^2 + 2x – 3$. (d) $x^2 + 2x + 3$.

12. The zeroes of the quadratic polynomial $x^2 + kx + k$, $k \neq 0$,
    (a) cannot both be positive (b) cannot both be negative
    (c) are always unequal (d) are always equal

MCQ WORKSHEET-II
CLASS X : CHAPTER - 2
POLYNOMIALS

1. If α, β are the zeroes of the polynomials $f(x) = x^2 + x + 1$, then $\frac{1}{\alpha}+\frac{1}{\beta}$

   (a) 0 (b) 1 (c) –1 (d) none of these

2. If one of the zero of the polynomial $f(x) = (k^2 + 4)x^2 + 13x + 4k$ is reciprocal of the other then k =

   (a) 2 (b) 1 (c) –1 (d) – 2

3. If α, β are the zeroes of the polynomials $f(x) = 4x^2 + 3x + 7$, then $\frac{1}{\alpha}+\frac{1}{\beta}$

   (a) $\frac{7}{3}$ (b) $\frac{-7}{3}$ (c) $\frac{3}{7}$ (d) $\frac{-3}{7}$

4. If the sum of the zeroes of the polynomial $f(x) = 2x^3 – 3kx^2 + 4x – 5$ is 6, then value of k is

   (a) 2 (b) 4 (c) –2 (d) – 4

5. The zeroes of a polynomial $p(x)$ are precisely the x-coordinates of the points, where the graph of $y = p(x)$ intersects the

   (a) x – axis (b) y – axis (c) origin (d) none of the above

6. If α, β are the zeroes of the polynomials $f(x) = x^2 – p(x + 1) – c$, then $(\alpha+1)(\beta+1)$ =

   (a) c – 1 (b) 1 – c (c) c (d) 1 + c

7. A quadratic polynomial can have at most …….. zeroes

   (a) 0 (b) 1 (c) 2 (d) 3

8. A cubic polynomial can have at most …….. zeroes.

   (a) 0 (b) 1 (c) 2 (d) 3

9. Which are the zeroes of $p(x) = x^2 – 1$:

   (a) 1, –1 (b) – 1, 2 (c) –2, 2 (d) –3, 3

10. Which are the zeroes of $p(x) = (x – 1)(x – 2)$:

    (a) 1, –2 (b) – 1, 2 (c) 1, 2 (d) –1, –2

11. Which of the following is a polynomial?

    $(a) x^2 - 5x + 3$

    $(b) \sqrt{x} + \frac{1}{\sqrt{x}}$

    $(c) x^{3/2} - x + x^{1/2}$

    $(d) x^{1/2} + x + 10$

12. Which of the following is not a polynomial?

    $(a) \sqrt{3}x^2 - 2\sqrt{3}x + 3$

    $(b) \frac{3}{2}x^3 - 5x^2 - \frac{1}{\sqrt{2}}x - 1$

    $(c) x + \frac{1}{x}$

    $(d) 5x^2 - 3x + \sqrt{2}$

MCQ WORKSHEET-III
CLASS X : CHAPTER - 2
POLYNOMIALS

1. If α, β are the zeroes of the polynomials $f(x) = x^2 + 5x + 8$, then $\alpha + \beta$
   (a) 5 (b) –5 (c) 8 (d) none of these

2. If α, β are the zeroes of the polynomials $f(x) = x^2 + 5x + 8$, then $\alpha.\beta$
   (a) 0 (b) 1 (c) –1 (d) none of these

3. On dividing $x^3 + 3x^2 + 3x + 1$ by $x + \pi$ we get remainder:
   $(a) -\pi^3 + 3\pi^2 - 3\pi + 1$
   $(b) \pi^3 - 3\pi^2 + 3\pi + 1$
   $(c) -\pi^3 - 3\pi^2 - 3\pi - 1$
   $(d) -\pi^3 + 3\pi^2 - 3\pi - 1$

4. The zero of $p(x) = 9x + 4$ is:
   (a) $\frac{4}{9}$ (b) $\frac{9}{4}$ (c) $\frac{-4}{9}$ (d) $\frac{-9}{4}$

5. On dividing $x^3 + 3x^2 + 3x + 1$ by $5 + 2x$ we get remainder:
   (a) $\frac{8}{27}$ (b) $\frac{-8}{27}$ (c) $\frac{-27}{8}$ (d) $\frac{27}{8}$

6. A quadratic polynomial whose sum and product of zeroes are –3 and 4 is
   (a) $x^2 - 3x + 12$ (b) $x^2 + 3x + 12$ (c) $2x^2 + x - 24$. (d) none of the above.

7. A quadratic polynomial whose zeroes are $\frac{3}{5}$ and $\frac{-1}{2}$ is
   (a) $10x^2 - x - 3$ (b) $10x^2 + x - 3$ (c) $10x^2 - x + 3$ (d) none of the above.

8. A quadratic polynomial whose sum and product of zeroes are 0 and 5 is
   (a) $x^2 - 5$ (b) $x^2 + 5$ (c) $x^2 + x - 5$. (d) none of the above.

9. A quadratic polynomial whose zeroes are 1 and –3 is
   (a) $x^2 - 2x - 3$ (b) $x^2 + 2x - 3$ (c) $x^2 - 2x + 3$ (d) none of the above.

10. A quadratic polynomial whose sum and product of zeroes are –5 and 6 is
    (a) $x^2 - 5x - 6$ (b) $x^2 + 5x - 6$ (c) $x^2 + 5x + 6$ (d) none of the above.

11. Which are the zeroes of $p(x) = x^2 + 3x - 10$:
    (a) 5, –2 (b) –5, 2 (c) –5, –2 (d) none of these

12. Which are the zeroes of $p(x) = 6x^2 - 7x - 3$:
    (a) 5, –2 (b) –5, 2 (c) –5, –2 (d) none of these

13. Which are the zeroes of $p(x) = x^2 + 7x + 12$:
    (a) 4, –3 (b) –4, 3 (c) –4, –3 (d) none of these

MCQ WORKSHEET-IV
CLASS X : CHAPTER - 2
POLYNOMIALS

1. The degree of the polynomial whose graph is given below:
(a) 1 (b) 2 (c) ≥ 3 (d) cannot be fixed

2. If the sum of the zeroes of the polynomial 3x2 – kx + 6 is 3, then the value of k is:
(a) 3 (b) –3 (c) 6 (d) 9

3. The other two zeroes of the polynomial x3 – 8x2 + 19x – 12 if tis one zeroes is x = 1 are:
(a) 3, –4 (b) –3, –4 (c) –3, 4 (d) 3, 4

4. The quadratic polynomial, the sum and product of whose zeroes are –3 and 2 is:
(a) $x^2 - 3x + 2$ (b) $x^2 + 3x - 2$ (c) $x^2 + 3x + 2$ (d) none of the these.

5. The third zero of the polynomial, if the sum and product of whose zeroes are –3 and 2 is:
(a) 7 (b) –7 (c) 14 (d) –14

6. If $\sqrt{\frac{5}{3}}$ and $-\sqrt{\frac{5}{3}}$ are two zeroes of the polynomial $3x^4 + 6x^3 - 2x^2 - 10x - 5$, then its other two zeroes are:
(a) –1, –1 (b) 1, –1 (c) 1, 1 (d) 3, –3

7. If a – b, a and a + b are zeroes of the polynomial $x^3 - 3x^2 + x + 1$ the value of (a + b) is
(a) $1 \pm \sqrt{2}$ (b) $-1 + \sqrt{2}$ (c) $-1 - \sqrt{2}$ (d) 3

8. A real numbers a is called a zero of the polynomial f(x), then
(a) f(a) = –1 (b) f(a) = 1 (c) f(a) = 0 (d) f(a) = –2

9. Which of the following is a polynomial:
(a) $x^2 + \frac{1}{x}$ (b) $2x^2 - 3\sqrt{x} + 1$ (c) $x^2 + x^{-2} + 7$ (d) $3x^2 - 3x + 1$

10. The product and sum of zeroes of the quadratic polynomial ax2 + bx + c respectively are:
(a) $\frac{b}{a}, \frac{c}{a}$ (b) $\frac{c}{a}, \frac{b}{a}$ (c) $\frac{c}{b}, 1$ (d) $\frac{c}{a}, \frac{-b}{a}$

11. The quadratic polynomial, sum and product of whose zeroes are 1 and –12 respectively is
(a) $x^2 - x - 12$ (b) $x^2 + x - 12$ (c) $x^2 - 12x + 1$ (d) $x^2 - 12x - 1$.

12. If the product of two of the zeroes of the polynomial $2x^3 - 9x^2 + 13x - 6$ is 2, the third zero of the polynomial is:
(a) –1 (b) –2 (c) $\frac{3}{2}$ (d) $-\frac{3}{2}$

PRACTICE QUESTIONS
CLASS X : CHAPTER - 2
POLYNOMIALS

1. If $p(x) = 3x^3 - 2x^2 + 6x - 5$, find $p(2)$.

2. Draw the graph of the polynomial $f(x) = x^2 - 2x - 8$.

3. Draw the graph of the polynomial $f(x) = 3 - 2x - x^2$.

4. Draw the graph of the polynomial $f(x) = -3x^2 + 2x - 1$.

5. Draw the graph of the polynomial $f(x) = x^2 - 6x + 9$.

6. Draw the graph of the polynomial $f(x) = x^3$.

7. Draw the graph of the polynomial $f(x) = x^3 - 4x$.

8. Draw the graph of the polynomial $f(x) = x^3 - 2x^2$.

9. Draw the graph of the polynomial $f(x) = -4x^2 + 4x - 1$.

10. Draw the graph of the polynomial $f(x) = 2x^2 - 4x + 5$.

11. Find the quadratic polynomial whose zeroes are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.

12. Find the quadratic polynomial whose zeroes are $\frac{3-\sqrt{3}}{5}$ and $\frac{3+\sqrt{3}}{5}$.

13. Find a quadratic polynomial whose sum and product of zeroes are $\sqrt{2}$ and 3 respectively.

14. Find the zeroes of the polynomial $mx^2 + (m + n)x + n$.

15. If m and n are zeroes of the polynomial $3x^2 + 11x - 4$, find the value of $\frac{m}{n} + \frac{n}{m}$

16. If a and b are zeroes of the polynomial $x^2 - x - 6$, then find a quadratic polynomial whose zeroes are $(3a + 2b)$ and $(2a + 3b)$.

17. If p and q are zeroes of the polynomial $t^2 - 4t + 3$, show that $\frac{1}{p} + \frac{1}{q} - 2pq + \frac{14}{3} = 0$.

18. If $(x - 6)$ is a factor of $x^3 + ax^2 + bx - b = 0$ and $a - b = 7$, find the values of a and b.

19. If 2 and -3 are the zeroes of the polynomial $x^2 + (a + 1)x + b$, then find the value of a and b.

20. Obtain all zeroes of polynomial $f(x) = 2x^4 + x^3 - 14x^2 - 19x - 6$ if two of its zeroes are -2 and -1.

21. Find all the zeroes of the polynomial $2x^3 - 4x - x^2 + 2$, if two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

22. Find all the zeroes of the polynomial $x^4 - 3x^3 + 6x - 4$, if two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

23. Find all the zeroes of the polynomial $2x^4 - 9x^3 + 5x^2 + 3x - 1$, if two of its zeroes are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.

24. Find all the zeroes of the polynomial $2x^4 + 7x^3 - 19x^2 - 14x + 30$, if two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

25. Find all the zeroes of the polynomial $x^3 + 3x^2 - 2x - 6$, if two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

26. Find all the zeroes of the polynomial $2x^3 - x^2 - 5x - 2$, if two of its zeroes are –1 and 2.

27. Find all the zeroes of the polynomial $x^3 + 3x^2 - 5x - 15$, if two of its zeroes are $\sqrt{5}$ and $-\sqrt{5}$.

28. Find all the zeroes of the polynomial $x^3 - 4x^2 - 3x + 12$, if two of its zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

29. Find all the zeroes of the polynomial $2x^3 + x^2 - 6x - 3$, if two of its zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

30. Find all the zeroes of the polynomial $x^4 + x^3 - 34x^2 - 4x + 120$, if two of its zeroes are 2 and –2.

31. If the polynomial $6x^4 + 8x^3 + 17x^2 + 21x + 7$ is divided by another polynomial $3x^2 + 4x + 1$, the remainder comes out to be (ax + b), find a and b.

32. If the polynomial $x^4 + 2x^3 + 8x^2 + 12x + 18$ is divided by another polynomial $x^2 + 5$, the remainder comes out to be px + q, find the value of p and q.

33. Find the zeroes of a polynomial $x^3 - 5x^2 - 16x + 80$, if its two zeroes are equal in magnitude but opposite in sign.

34. If two zeroes of the polynomial $x^4 + 3x^3 - 20x^2 - 6x + 36$ are $\sqrt{2}$ and $-\sqrt{2}$, find the other zeroes of the polynomial.

35. On dividing $x^3 - 3x^2 + x + 2$ by a polynomial g(x), the quotient and remainder were x – 2 and –2x + 4 respectively. Find g(x).

36. If the product of zeroes of the polynomial $ax^2 - 6x - 6$ is 4, find the value of 'a'.

37. If one zero of the polynomial $(a^2 + 9)x^2 + 13x + 6a$ is reciprocal of the other. Find the value of a.

38. Write a quadratic polynomial, sum of whose zeroes is $2\sqrt{3}$ and their product is 2.

39. Find a polynomial whose zeroes are 2 and –3.

40. Find the zeroes of the quadratic polynomial $x^2 + 5x + 6$ and verify the relationship between the zeroes and the coefficients.

41. Find the sum and product of zeroes of $p(x) = 2(x^2 - 3) + x$.

42. Find a quadratic polynomial, the sum of whose zeroes is 4 and one zero is 5.

43. Find the zeroes of the polynomial $p(x) = \sqrt{2}x^2 - 3x - 2\sqrt{2}$.

44. If α and β are the zeroes of $2x^2 + 5(x - 2)$, then find the product of α and β.

45. Find a quadratic polynomial, the sum and product of whose zeroes are 5 and 3 respectively.

46. Find the zeroes of the quadratic polynomial $f(x) = abx^2 + (b^2 - ac)x - bc$ and verify the relationship between the zeroes and its coefficients.

47. Find the zeroes of the following polynomials by factorisation method and verify the relations between the zeroes and the coefficients of the polynomials:

(i) $4x^2 - 3x - 1$

(ii) $3x^2 + 4x - 4$

(iii) $5t^2 + 12t + 7$

(iv) $t^3 - 2t^2 - 15t$

(v) $2x^2 + \frac{7}{2}x + \frac{3}{4}$

(vi) $4x^2 + 5\sqrt{2}\,x - 3$

(vii) $2s^2 - (1 + 2\sqrt{2})s + \sqrt{2}$

(viii) $v^2 + 4\sqrt{3}\,v - 15$

(ix) $y^2 + \frac{3}{2}\sqrt{5}\,y - 5$

(x) $7y^2 - \frac{11}{3}y - \frac{2}{3}$

48. Find the zeroes of the quadratic polynomial $6x^2 - 7x - 3$ and verify the relationship between the zeroes and the coefficients.

49. Find the zeroes of the polynomial $x^2 + \frac{1}{6}x - 2$, and verify the relation between the coefficients and the zeroes of the polynomial.

50. Find the zeroes of the quadratic polynomial $x^2 + 5x + 6$ and verify the relationship between the zeroes and the coefficients.

51. Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and $-\frac{3}{2}$, respectively. Also find its zeroes.

52. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then find the value of *k*

53. Given that two of the zeroes of the cubic polynomial $ax^3 + bx^2 + cx + d$ are 0, find the third zero.

54. Given that one of the zeroes of the cubic polynomial $ax^3 + bx^2 + cx + d$ is zero, then find the product of the other two zeroes.

55. If one of the zeroes of the cubic polynomial $x^3 + ax^2 + bx + c$ is –1, then the product of the other two zeroes

Answer the Questions from 28 to 32 and justify:

56. Can $x^2 - 1$ be the quotient on division of $x^6 + 2x^3 + x - 1$ by a polynomial in *x* of degree 5?

57. What will the quotient and remainder be on division of $ax^2 + bx + c$ by $px^3 + qx^2 + rx + s$, $p \neq 0$?

58. If on division of a polynomial $p(x)$ by a polynomial $g(x)$, the degree of quotient is zero, what is the relation between the degrees of $p(x)$ and $g(x)$?

59. If on division of a non-zero polynomial $p(x)$ by a polynomial $g(x)$, the remainder is zero, what is the relation between the degrees of $p(x)$ and $g(x)$?

60. Can the quadratic polynomial $x^2 + kx + k$ have equal zeroes for some odd integer $k > 1$?

61. If one of the zeroes of the quadratic polynomial $(k-1)x^2 + k x + 1$ is -3, then the value of k

62. If the zeroes of the quadratic polynomial $x^2 + (a + 1)x + b$ are 2 and -3, then find the value of a and b.

63. If α and β are zeroes of the quadratic polynomial $x^2 - (k + 6)x + 2(2k - 1)$. Find the value of k if $\alpha + \beta = \frac{1}{2}\alpha\beta$.

64. Obtain all the zeroes of $3x^4 + 6x^3 - 2x^2 - 10x + 5$, if two of its zeroes are $\sqrt{\frac{5}{3}}$ and $-\sqrt{\frac{5}{3}}$.

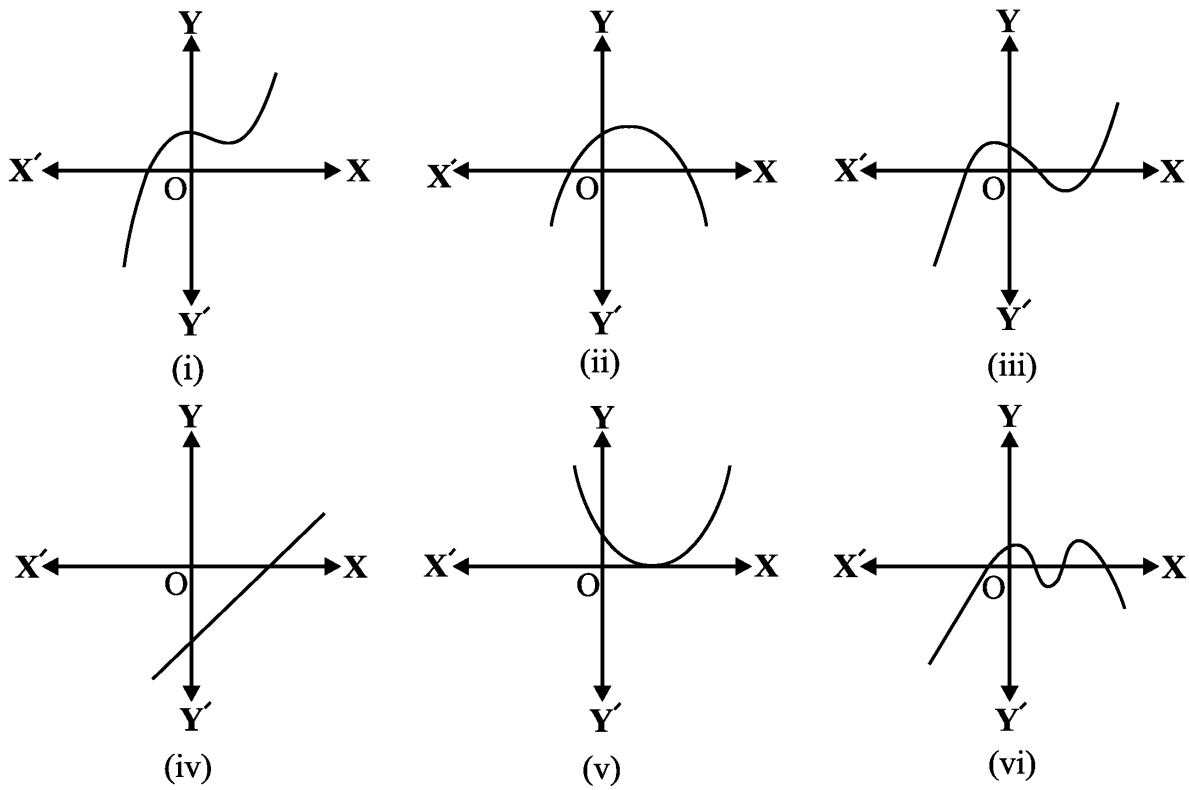
65. Obtain all the zeroes of $x^4 - 7x^3 + 17x^2 - 17x + 6$, if two of its zeroes are 3 and 1.

66. Obtain all the zeroes of $x^4 - 7x^2 + 12$, if two of its zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

67. Two zeroes of the cubic polynomial $ax^3 + 3x^2 - bx - 6$ are -1 and -2. Find the 3rd zero and value of a and b.

68. α, β and γ are the zeroes of cubic polynomial $x^3 + px^2 + qx + 2$ such that $\alpha . \beta + 1 = 0$. Find the value of $2p + q + 5$.

69. Find the number of zeroes in each of the following:



70. If the remainder on division of $x^3 + 2x^2 + kx + 3$ by $x - 3$ is 21, find the quotient and the value of k. Hence, find the zeroes of the cubic polynomial $x^3 + 2x^2 + kx - 18$.

71. Find the zeroes of the polynomial $f(x) = x^3 - 5x^2 - 16x + 80$, if its two zeroes are equal in magnitude but opposite in sign.

72. Find the zeroes of the polynomial $f(x) = x^3 - 5x^2 - 2x + 24$, if it is given that the product of two zeroes is 12.

73. Find the zeroes of the polynomial $f(x) = x^3 - px^2 + qx - r$, if it is given that the sum of two zeroes is zero.

74. If the zeroes of the polynomial $x^3 - 3x^2 + x + 1$ are $a - b$, a, $a + b$, find a and b.

75. If the zeroes of the polynomial $2x^3 - 15x^2 + 37x - 30$ are $a - b$, a, $a + b$, find all the zeroes.

76. If the zeroes of the polynomial $x^3 - 12x^2 + 39x - 28$ are $a - b$, a, $a + b$, find all the zeroes.

77. If the polynomial $x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $x^2 - 2x + k$, the remainder comes out to be $x + a$, find k and a.

78. If the polynomial $6x^4 + 8x^3 - 5x^2 + ax + b$ is exactly divisible by the polynomial $2x^2 - 5$, then find the values of a and b.

79. Find a cubic polynomial with the sum, sum of the product of its zeroes taken two at a time, and the product of its zeroes as 2, –7, –14 respectively.

80. Find a cubic polynomial with the sum, sum of the product of its zeroes taken two at a time, and the product of its zeroes as 3, –1, –3 respectively.

81. Find a cubic polynomial whose zeroes are 3, $\frac{1}{2}$ and –1.

82. Find a cubic polynomial whose zeroes are –2, –3 and –1.

83. Find a cubic polynomial whose zeroes are 3, 5 and –2.

84. Verify that 5, –2 and $\frac{1}{3}$ are the zeroes of the cubic polynomial $p(x) = 3x^3 - 10x^2 - 27x + 10$ and verify the relation between its zeroes and coefficients.

85. Verify that 3, –2 and 1 are the zeroes of the cubic polynomial $p(x) = x^3 - 2x^2 - 5x + 6$ and verify the relation between its zeroes and coefficients.

86. Verify that the numbers given alongside of the cubic polynomials below are their zeroes. Also verify the relationship between the zeroes and the coefficients in each case:

(i) $2x^3 + x^2 - 5x + 2$; $\frac{1}{2}$, 1, – 2 (ii) $x^3 - 4x^2 + 5x - 2$; 2, 1, 1

87. Find the quotient and remainder when $4x^3 + 2x^2 + 5x - 6$ is divided by $2x^2 + 3x + 1$.

88. On dividing $x^4 - 5x + 6$ by a polynomial $g(x)$, the quotient and remainder were $-x^2 - 2$ and $-5x + 10$ respectively. Find $g(x)$.

89. Given that $\sqrt{2}$ is a zero of the cubic polynomial $6x^3 + \sqrt{2}\,x^2 - 10x - 4\sqrt{2}$, find its other two zeroes.

90. Given that the zeroes of the cubic polynomial $x^3 - 6x^2 + 3x + 10$ are of the form a, $a + b$, $a + 2b$ for some real numbers a and b, find the values of a and b as well as the zeroes of the given polynomial.

91. For which values of a and b, are the zeroes of $q(x) = x^3 + 2x^2 + a$ also the zeroes of the polynomial $p(x) = x^5 - x^4 - 4x^3 + 3x^2 + 3x + b$? Which zeroes of $p(x)$ are not the zeroes of $q(x)$?

92. Find k so that $x^2 + 2x + k$ is a factor of $2x^4 + x^3 - 14x^2 + 5x + 6$. Also find all the zeroes of the two polynomials.

93. Given that $x - \sqrt{5}$ is a factor of the cubic polynomial $x^3 - 3\sqrt{5}x + 13x - 3\sqrt{5}$, find all the zeroes of the polynomial.

94. For each of the following, find a quadratic polynomial whose sum and product respectively of the zeroes are as given. Also find the zeroes of these polynomials by factorisation.

$(i)\ \frac{-8}{3}, \frac{4}{3}$ $(ii)\ \frac{21}{8}, \frac{5}{16}$

$(iii)\ -2\sqrt{3}, -9$ $(iv)\ \frac{-3}{2\sqrt{5}}, -\frac{1}{2}$

95. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 3x - 2$, then find a quadratic polynomial whose zeroes are $\frac{1}{2\alpha+\beta}$ and $\frac{1}{2\beta+\alpha}$.

96. If α and β are the zeroes of the quadratic polynomial $f(x) = 2x^2 - 5x + 7$, then find a quadratic polynomial whose zeroes are $2\alpha+3\beta$ and $2\beta+3\alpha$.

97. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 1$, then find a quadratic polynomial whose zeroes are $\frac{2\alpha}{\beta}$ and $\frac{2\beta}{\alpha}$.

98. If α and β are the zeroes of the quadratic polynomial $f(x) = 6x^2 + x - 2$, then find the value of

$(i)\ \alpha - \beta$ $(ii)\ \alpha^2 + \beta^2$ $(iii)\ \alpha^4 + \beta^4$ $(iv)\ \alpha\beta^2 + \alpha^2\beta$

$(v)\ \frac{1}{\alpha} + \frac{1}{\beta}$ $(vi)\ \frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$ $(vii)\ \frac{1}{\alpha} - \frac{1}{\beta}$ $(viii)\ \alpha^3 + \beta^3$

$(ix)\ \frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ $(x)\ \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ $(xi)\ \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$

$(xii)\ \alpha^4\beta^3 + \alpha^3\beta^4$ $(xiii)\ \frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$ $(xiv)\ \frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2}$

99. If α and β are the zeroes of the quadratic polynomial $f(x) = 4x^2 - 5x - 1$, then find the value of

$(i)\ \alpha - \beta$ $(ii)\ \alpha^2 + \beta^2$ $(iii)\ \alpha^4 + \beta^4$ $(iv)\ \alpha\beta^2 + \alpha^2\beta$

$(v)\ \frac{1}{\alpha} + \frac{1}{\beta}$ $(vi)\ \frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$ $(vii)\ \frac{1}{\alpha} - \frac{1}{\beta}$ $(viii)\ \alpha^3 + \beta^3$

$(ix)\ \frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ $(x)\ \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ $(xi)\ \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$

$(xii)\ \alpha^4\beta^3 + \alpha^3\beta^4$ $(xiii)\ \frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$ $(xiv)\ \frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2}$

100. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 + x - 2$, then find the value of

$(i)\alpha - \beta$ $(ii)\alpha^2 + \beta^2$ $(iii)\alpha^4 + \beta^4$ $(iv)\alpha\beta^2 + \alpha^2\beta$

$(v)\frac{1}{\alpha} + \frac{1}{\beta}$ $(vi)\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$ $(vii)\frac{1}{\alpha} - \frac{1}{\beta}$ $(viii)\alpha^3 + \beta^3$

$(ix)\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ $(x)\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ $(xi)\frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$

$(xii)\alpha^4\beta^3 + \alpha^3\beta^4$ $(xiii)\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$ $(xiv)\frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2}$

101. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 5x + 4$, then find the value of

$(i)\alpha - \beta$ $(ii)\alpha^2 + \beta^2$ $(iii)\alpha^4 + \beta^4$ $(iv)\alpha\beta^2 + \alpha^2\beta$

$(v)\frac{1}{\alpha} + \frac{1}{\beta}$ $(vi)\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$ $(vii)\frac{1}{\alpha} - \frac{1}{\beta}$ $(viii)\alpha^3 + \beta^3$

$(ix)\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ $(x)\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ $(xi)\frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$

$(xii)\alpha^4\beta^3 + \alpha^3\beta^4$ $(xiii)\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$ $(xiv)\frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2}$

102. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 2x + 3$, then find a quadratic polynomial whose zeroes are $\alpha + 2$ and $\beta + 2$

103. If α and β are the zeroes of the quadratic polynomial $f(x) = 3x^2 - 4x + 1$, then find a quadratic polynomial whose zeroes are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$.

104. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 2x + 3$, then find a quadratic polynomial whose zeroes are $\frac{\alpha - 1}{\alpha + 1}$ and $\frac{\beta - 1}{\beta + 1}$.

105. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - p(x + 1) - c$, show that $(\alpha + 1)(\beta + 1) = 1 - c$.

106. If α and β are the zeroes of the quadratic polynomial such that $\alpha + \beta = 24$ and $\alpha - \beta = 8$, find a quadratic polynomial having α and β as its zeroes.

107. If sum of the squares of zeroes of the quadratic polynomial $f(x) = x^2 - 8x + k$ is 40, find the value of k.

108. If α and β are the zeroes of the quadratic polynomial $f(x) = kx^2 + 4x + 4$ such that $\alpha^2 + \beta^2 = 24$, find the value of k.

109. If α and β are the zeroes of the quadratic polynomial $f(x) = 2x^2 + 5x + k$ such that $\alpha^2 + \beta^2 + \alpha\beta = \frac{21}{4}$, find the value of k.

110. What must be subtracted from $8x^4 + 14x^3 - 2x^2 + 7x - 8$ so that the resulting polynomial is exactly divisible by $4x^2 + 3x - 2$.

111. What must be subtracted from $4x^4 + 2x^3 - 2x^2 + x - 1$ so that the resulting polynomial is exactly divisible by $x^2 + 2x - 3$.

112. Find all the zeroes of the polynomial $x^4 - 6x^3 - 26x^2 + 138x - 35$, if two of its zeroes are $2+\sqrt{3}$ and $2-\sqrt{3}$.

113. Find the values of a and b so that $x^4 + x^3 + 8x^2 + ax + b$ is divisible by $x^2 + 1$.

114. If the polynomial $f(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $x^2 - 2x + k$, the remainder comes out to be $x + a$, find k and a.

115. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 2x - 8$, then find the value of

(i) $\alpha - \beta$ $(ii)\ \alpha^2 + \beta^2$ $(iii)\ \alpha^4 + \beta^4$ $(iv)\ \alpha\beta^2 + \alpha^2\beta$

$(v)\ \frac{1}{\alpha} + \frac{1}{\beta}$ $(vi)\ \frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$ $(vii)\ \frac{1}{\alpha} - \frac{1}{\beta}$ $(viii)\ \alpha^3 + \beta^3$

$(ix)\ \frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ $(x)\ \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ $(xi)\ \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$

$(xii)\ \alpha^4\beta^3 + \alpha^3\beta^4$ $(xiii)\ \frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$ $(xiv)\ \frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2}$